INCH-POUND

MIL-PRF-27/320C
AMENDMENT 1
27 July 2017
SUPERSEDING
MIL-PRF-27/320C
28 April 2008

PERFORMANCE SPECIFICATION SHEET

# TRANSFORMERS, POWER, 160 VOLT-AMPERES, 60 HERTZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

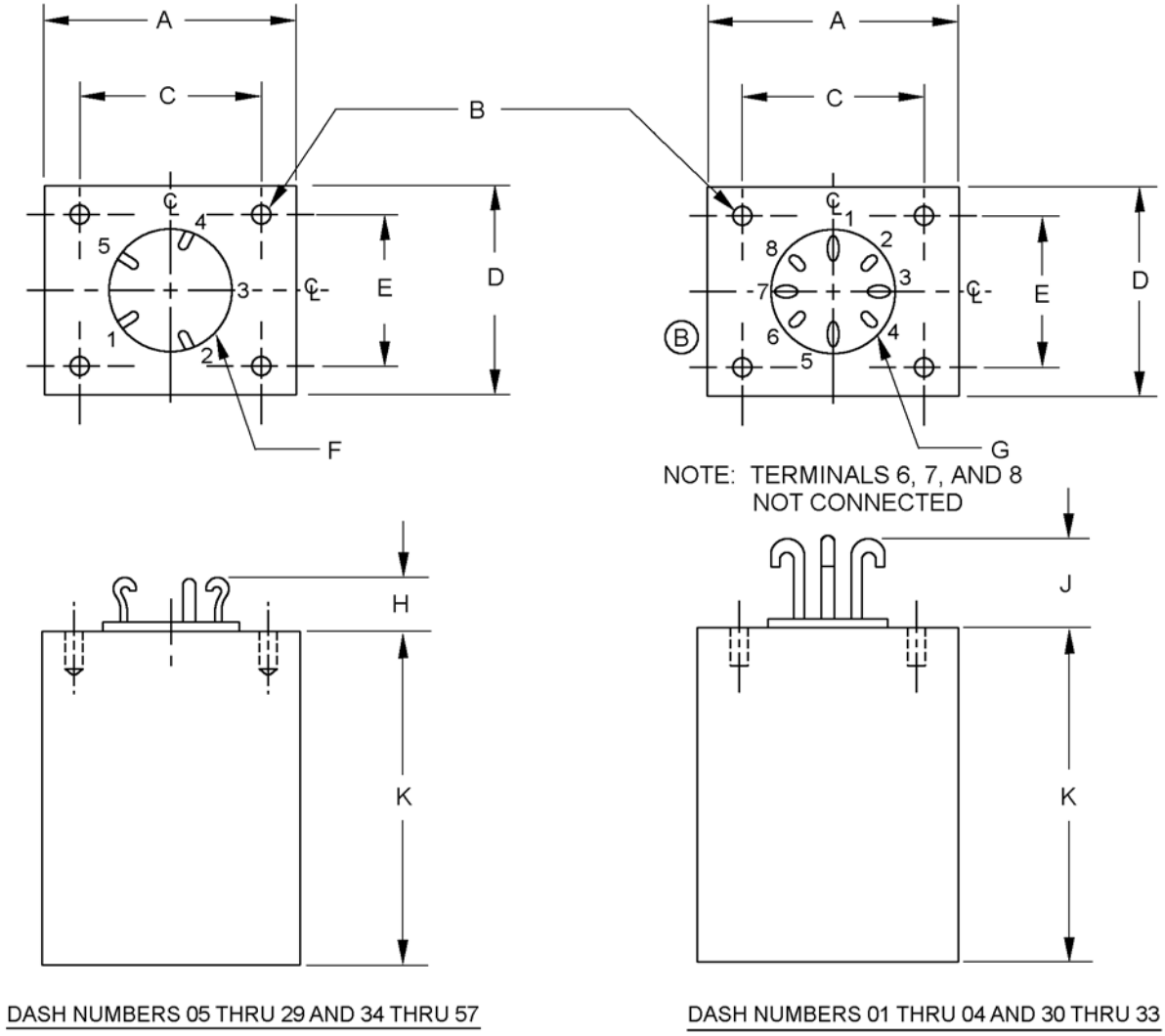
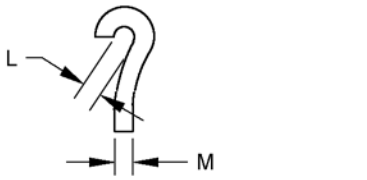


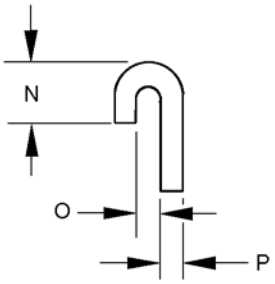

FIGURE 1. Dimensions and configurations.



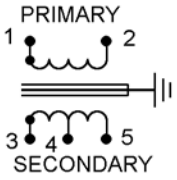
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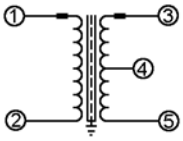

DASH NUMBER 05 THRU 29 AND 34 THRU 57




DASH NUMBERS 01 THRU 04 AND 30 THRU 33




CIRCUIT DIAGRAM "A"




CIRCUIT DIAGRAM "B"

| Dim | Inches | mm | Dim | Inches | mm |
|---|---|---|---|---|---|
| A | 3.75 ±.031 | ±0.79 | J | .470 ±.031 | 11.94 ±0.79 |
| B | .190-32 UNF-2B (4 PLACES) 8 FULL THDS min. | | K | 4.88 | 123.95 |
| C | 2.75 ±.016 | ±0.41 | L | .070 ±.005 | 1.78 ±0.13 |
| D | 3.13 ±.031 | ±0.79 | M | .062 ±.005 | 1.57 ±0.13 |
| E | 2.44 ±.016 | ±0.41 | N | .188 ±.005 | 4.78 ±0.13 |
| F | .875 DIA | 22.22 DIA | O | .094 ±.005 | 2.39 ±0.13 |
| G | 1.063 DIA | 27.00 DIA | P | .080 ±.005 | 2.03 ±0.13 |
| H | .375 ±.031 | 9.52 ±0.79 | | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the sides of the case.
4. Unless otherwise specified, tolerance shall be ±.063 (1.60 mm).
5. Electrical values shall be marked as specified in table I, except DCR.

FIGURE 1. Dimensions and configurations.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 115 volts ac, 50 to 60 hertz.

Primary dc resistance: 1.6 ohms ±25 percent.

Secondary voltage (3-5): See table I.

Volt-ampere rating: 160 volt-amperes.

Working voltage (1-2): 535 volts peak.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle. Continuous.

Case: Metal enclosed.

Material: Steel.

Terminals: Solderable terminals.

| Dash numbers | Terminal type | Height (inches) | Diameter (inches) |
|---|---|---|---|
| 01 through 04 | Hook | 0.470 ±.031 | 0.080 ±.005 |
| 05 through 29 | Hook | 0.375 ±.031 | 0.062 ±.005 |
| 30 through 33 | Hook | 0.470 ±.031 | 0.080 ±.005 |
| 34 through 56 | Hook | 0.375 ±.031 | 0.062 ±.005 |

Weight: 8.2 pounds, maximum.

Operating temperature range. -55° to +130°C.

Altitude: 10,000 feet.

Terminal strength: MIL-STD-202-211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (each winding) (at sea level): 1,500 volts rms.

Electrical characteristics:

Rated load: With 115 volts ac and 60 Hz across (1-2), and rated current in secondary, the voltage across (3-5) shall be as specified in table I.

Regulation: $\frac{Voltage\,(no\ load) - Voltage\,(rated\ load)}{Voltage\,(rated\ load)}\ x\ 100$

Shall not exceed 10 percent.

Polarity: Additive with terminals 2 and 3 connected.

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Temperature rise: 35°C with 115 volts ac, 60 hertz across (1-2) at an ambient temperature of 95°C, full load terminals (3-5).

Marking location: See figure 1.

Part or Identification Number (PIN): M27/320-(dash number from table I).

TABLE I. Electrical ratings.

| Dash number | Secondary (3-5) | | | |
|---|---|---|---|---|
| | Voltage (volts) ±3 percent | Current 1/ (amperes) | DC resistance (ohms) ±25 percent | Circuit diagram |
| 01 | 8.0 | 20.00 | .014 | A |
| 02 | 10.0 | 16.00 | .018 | A |
| 03 | 12.6 | 12.70 | .038 | A |
| 04 | 14.0 | 11.40 | .042 | A |
| 05 | 16.0 | 10.00 | .046 | A |
| 06 | 18.0 | 8.89 | .060 | A |
| 07 | 20.0 | 8.00 | .070 | A |
| 08 | 22.0 | 7.27 | .092 | A |
| 09 | 25.0 | 6.40 | 0.110 | A |
| 10 | 28.0 | 5.72 | 0.150 | A |
| 11 | 31.0 | 5.17 | 0.166 | A |
| 12 | 35.0 | 4.57 | 0.240 | A |
| 13 | 39.0 | 4.10 | 0.270 | A |
| 14 | 43.0 | 3.70 | 0.382 | A |
| 15 | 48.0 | 3.33 | 0.405 | A |
| 16 | 55.0 | 2.91 | 0.600 | A |
| 17 | 63.0 | 2.52 | 0.640 | A |
| 18 | 70.0 | 2.280 | 0.93 | A |
| 19 | 80.0 | 2.000 | 1.06 | A |
| 20 | 90.0 | 1.780 | 1.49 | A |
| 21 | 100 | 1.600 | 1.69 | A |
| 22 | 110 | 1.450 | 2.30 | A |
| 23 | 123 | 1.300 | 2.57 | A |
| 24 | 138 | 1.150 | 3.60 | A |
| 25 | 157 | 1.020 | 4.10 | A |
| 26 | 175 | 0.914 | 5.82 | A |
| 27 | 200 | 0.800 | 6.70 | A |
| 28 | 220 | 0.728 | 8.95 | A |

See footnote at end of table.

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TABLE I. Electrical ratings - Continued.

| Dash number | Secondary (3-5) | | | |
|---|---|---|---|---|
| | Voltage (volts) ±3 percent | Current 1/ (amperes) | DC resistance (ohms) ±25 percent | Circuit diagram |
| 29 | 40.0 | 4.10 | 0.277 | B |
| 30 | 8.0 | 20.00 | .014 | B |
| 31 | 10.0 | 16.00 | .018 | B |
| 32 | 12.6 | 12.70 | .038 | B |
| 33 | 14.0 | 11.40 | .042 | B |
| 34 | 16.0 | 10.00 | .046 | B |
| 35 | 18.0 | 8.89 | .060 | B |
| 36 | 20.0 | 8.00 | .070 | B |
| 37 | 22.0 | 7.27 | .092 | B |
| 38 | 25.0 | 6.40 | 0.110 | B |
| 39 | 28.0 | 5.72 | 0.150 | B |
| 40 | 31.0 | 5.17 | 0.166 | B |
| 41 | 35.0 | 4.57 | 0.240 | B |
| 42 | 39.0 | 4.10 | 0.270 | B |
| 43 | 43.0 | 3.70 | 0.382 | B |
| 44 | 48.0 | 3.33 | 0.405 | B |
| 45 | 55.0 | 2.91 | 0.600 | B |
| 46 | 63.0 | 2.52 | 0.640 | B |
| 47 | 70.0 | 2.280 | 0.93 | B |
| 48 | 80.0 | 2.000 | 1.06 | B |
| 49 | 90.0 | 1.780 | 1.49 | B |
| 50 | 100 | 1.600 | 1.69 | B |
| 51 | 110 | 1.450 | 2.30 | B |
| 52 | 123 | 1.300 | 2.57 | B |
| 53 | 138 | 1.150 | 3.60 | B |
| 54 | 157 | 1.020 | 4.10 | B |
| 55 | 175 | 0.914 | 5.82 | B |
| 56 | 200 | 0.800 | 6.70 | B |
| 57 | 220 | 0.728 | 8.95 | B |

1/ For 50 Hz operation, reduce secondary current by 10 percent.

VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/312-26 or M27/316-30, and M27/322-24 shall be sufficient to grant qualification approval to MIL-PRF-27/312 through MIL-PRF-27/322, inclusive, all parts.

Qualification testing and approval to M27/320-28 shall be sufficient to grant qualification approval M27/320-01 through M27/320-56.

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-211

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2017-022)

Review activities:
Army – AR, CR4
Navy - AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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